Research article

A time-controllable Allan variance method for MEMS IMU

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Abstract

Purpose – The purpose of this paper is to reduce the calculation burden and speed up the estimation process of Allan variance method while ensuring the exactness of the analysis results.

Design/methodology/approach – A series of six-hour static tests have been implemented at room temperature, and the static measurements have been collected from MEMS IMU. In order to characterize the various types of random noise terms for the IMU, the basic definition and main procedure of the Allan variance method are investigated. Unlike the normal Allan variance method, which has the shortcomings of processing large data sets and requiring long computation time, a modified Allan variance method is proposed based on the features of data distribution in the log-log plot of the Allan standard deviation versus the averaging time.

Findings – Experiment results demonstrate that the modified Allan variance method can effectively estimate the noise coefficients for MEMS IMU, with controllable computation time and acceptable estimation accuracy.

Originality/value – This paper proposes a time-controllable Allan variance method which can quickly and accurately identify different noise terms imposed by the stochastic fluctuations.

Keywords Stability (control theory), Noise, Allan variance, Micro-electro-mechanical systems, Inertial measurement unit, MEMS IMU, Noise coefficient, Gyroscope

Paper type Research paper

I. Introduction

Inertial navigation system (INS) is a self-contained, non-radiating and anti-jamming navigation system, which provides dynamic information through direct measurements of inertial sensors. INS does not require extensive infrastructure pre-installed around or above a tracking environment, where the installation is time-consuming, expensive and completely unfeasible in the case of emergency response. Thus, INS is appealing in view of the lack of dependence on artificially generated sources.

However, INS is exceedingly sensitive to drift. For dead-reckoning purpose, the output of IMU must be numerically integrated to provide orientation, velocity and position information. Although the output is reliable over long periods of time, any small error will accumulate and grow without bound due to this successive integration with respect to time, especially for the low-cost MEMS IMU. Therefore, the reliability and accuracy of INS is questionable. One way of eliminating the accumulated errors is to periodically recalibrate inertial sensors with other absolute position measurements. Thus, in most situations INS needs to be integrated with other absolute positioning system to decrease the long-term navigation errors (Kelly and Sukhatme, 2011; Ryu and Gerdes, 2004). Until recently, the scientific literature only offers very few implementations of INS stand-alone system.

Another disadvantage of INS is the high cost of inertial sensors. In the early days of INSs, they had been widely used

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in aerospace applications (Barshan and Durrant-Whyte, 1995). The high-quality inertial sensors developed for aerospace are far too expensive for the general consumers due to budget constraints. This situation gradually changed with the reduction in the cost of inertial sensors, for example, the low-cost MEMS IMUs. Due to recent breakthrough developments in micro-machining technology, the costs of micro-machined accelerometers are decreasing while the accuracy is being improved. However, the micro-machined low-cost gyroscopes cannot achieve the required accuracy for precise navigation applications (Parsa *et al.*, 2007). Typically, MEMS gyroscopes have large bias drifts, and these drifts can accumulate several degrees of orientation error after even 1 min. Such large error rates make it difficult to use suitably priced gyroscopes in navigation applications.

The use of MEMS IMU in consumer-based applications has increased in the past decade, such as MEMS-based wearable motion capture system (Brigante *et al.*, 2011), low-cost personal dead-reckoning system (Abdulrahim *et al.*, 2011; Huang *et al.*, 2010), IMU aided SLAM system (Kelly and Sukhatme, 2011; Martinelli, 2012) and INS/GPS navigation system (Campbell, 2012; North *et al.*, 2009; Qin *et al.*, 2012). Taking the examples of INS/SLAM and INS/GPS integrated systems, the Kalman filter (KF) and its variations have been implemented as the primary integration scheme (Kim and Sukkarieh, 2007; Noureldin *et al.*, 2009). The first steps of KF implementation are the stochastic error modeling and the design of process noise covariance matrix Q in the prediction phase (Almagbile *et al.*, 2010; Ding *et al.*, 2007). Therefore, error estimation and error modeling are essential for the performance of INS, and a series of experiments need to be performed to characterize the errors of IMUs before putting them into practical use. With proper calibration and error modeling, the positioning and navigation accuracy can be greatly improved (Aggarwal *et al.*, 2008; Wang *et al.*, 2010).

The stochastic errors of inertial sensors can be modeled using either the time-domain technique or the frequency-domain technique. Each of them has its own advantages and disadvantages (El-Sheimy *et al.*, 2008; Hou, 2004). The frequency-domain technique uses the power spectral density (PSD) to estimate transfer functions, which is straightforward, but the computation is complex and difficult for non-system analysts to understand. In the time-domain technique, the correlation method is being used for analyzing sensor measurements. However, the correlation method is very model sensitive and needs long-term signals for an acceptable solution. Although several variance methods in the time-domain have been devised, they are basically very similar and primarily differ only in various signal processing methods. In comparison with other stochastic modeling methods, the Allan variance method is the simplest time-domain method, which is widely used to characterize the stochastic noise in MEMS sensors (Dean *et al.*, 2011; Hidalgo *et al.*, 2012; Zhang *et al.*, 2008).

This paper starts with an investigation of the theoretical basis and implementation process of the Allan variance method while identifying various noise terms existing in MEMS gyroscopes, and then displays the shortcomings of the normal Allan variance method such as processing large data set and requiring long computation time. However, the published work rarely mentions the computation time of Allan variance method, which is especially crucial for long-term sample data, e.g. 18 hours of static data (Aggarwal *et al.*, 2008). The purpose of this paper is to reduce the calculation burden and speed up the estimation process of Allan variance method while ensuring the exactness of the results. A modified time-controllable Allan variance method is proposed, and long-term static tests demonstrate that the modified method could greatly improve the computation speed and ensure the accuracy of estimation.

II. Allan variance method

As a researcher at NIST, Dr David Allan developed a method for characterizing random noise terms and stability in oscillators used in clock applications, which is known as the Allan (1966) variance method. Although developed for clock systems, Allan variance method can be a useful means to investigate the time-domain characteristics of random noise terms contained in inertial sensor data (IEEE, 1999, 2006). Even for the low-cost MEMS inertial sensors, test results showed that their noise terms can still be identified and characterized by the Allan variance method (El-Sheimy *et al.*, 2008; Hou and El-Sheimy, 2003).

A. Methodology

Assuming the sampling interval is T_s and the total sampling time is T, the number of data points contained in the entire data set can be obtained by $N = T/T_s$. The entire data set is divided into clusters of n consecutive data points, where n ($n = 1, 2, \ldots, n_{MAX}$, $n_{MAX} < N/2$) is the cluster size, and n_{MAX} is the maximum cluster size allowed for Allan variance calculation. $K = N/n$ is defined as the number of divided independent clusters, so $K_{MIN} = N/n_{MAX}$ is the minimum number of clusters.

At present, most commercially produced MEMS gyroscopes are angular rate gyroscopes. Therefore, only the angular rate gyroscopes are considered in this paper. If the instantaneous output of an angular rate gyroscope is $\omega(t)$, the averaging operation over the $(k+1)$th cluster is defined as:

$$\bar{\omega}_{k+1}(\tau) = \frac{1}{n}\sum_{i=1}^{n} \omega_{nk+i}, \tag{1}$$

where $\tau = nT_s$ is the averaging time of each cluster.

Then, the Allan variance can be defined based on the averages of angular velocity as:

$$\sigma^2(\tau) = \frac{1}{2(K-1)}\sum_{k=1}^{K-1}\left((\bar{\omega}_{k+1}(\tau) - \bar{\omega}_k(\tau))^2\right). \tag{2}$$

Since the Allan variance is a measure of the bias stability of inertial sensors, there is an integration relationship between $\sigma^2(\tau)$ and the PSD of the intrinsic stochastic processes as:

$$\sigma^2(\tau) = 4\int_0^{\infty} S_\omega(f)\frac{\sin^4(\pi f\tau)}{(\pi f\tau)^2}\,df, \tag{3}$$

where $S_\omega(f)$ is the PSD of stochastic process $\omega(t)$ (IEEE, 2006; Tehrani, 1983).

Equation (3) indicates that the Allan variance is proportional to the total power output of the stochastic process when passing through a filter with a transfer function in the form of $\sin^4 x/x^2$ and with a filter bandwidth depending on τ. Therefore, by adjusting the filter bandwidth, namely by varying τ, various types of stochastic noise terms can be

identified and quantified. Normally, a log-log plot of $\sigma(\tau)$ versus τ can provide a direct indication of the types of stochastic noise terms that exist in the inertial sensor data, where $\sigma(\tau)$ is the square root of the Allan variance, also referred to as the Allan standard deviation.

B. Noise terms

There are a number of stochastic noise terms, which are either known to exist in the inertial sensor or suspected to influence the data (Hou, 2004). Totally, there are seven noise terms related to the Allan's definition and results, and a typical log-log plot of $\sigma(\tau)$ versus τ should have the similar characteristic curve as shown in Figure 1. The characteristic curve can be viewed as an end-to-end connection of several segments with different slopes.

Generally, any of the seven noise terms can be present in the sample data and identified through the Allan variance method. Academic research and practical experience indicate that, in most cases, different noise terms appear in different regions of τ with different slopes, which allows easy identification of the stochastic noise terms existing in the data. Normally, for the lower accuracy inertial sensor, its Allan deviation curve is closer to the deviation axis and farther from the time axis; while the opposite is true for the higher-grade inertial sensor. In particular, for a certain inertial sensor, its Allan characteristic curve may not have the typical global structure shown in Figure 1, which may only have one or a few basic segments instead. This is because some of the noise terms do not exist or are too small to distinguish from other noise terms with greater Allan deviation slopes. These noise terms will not be present until the major noise terms are substantially compensated or reduced.

C. Estimation accuracy

In practical application, Allan variance estimation is based on a finite number of independent clusters that can be formed from any data set with finite length. The parameter $\sigma(\delta_{AV})$ is defined as the percentage error in estimating the Allan standard deviations of the independent clusters. Due to the finiteness of the divided clusters, the percentage error can be obtained by a straightforward calculation as (Papoulis, 1991):

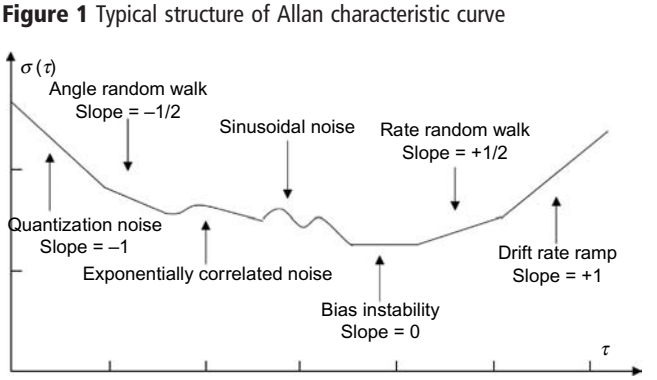
$$\sigma(\delta_{AV}) = \frac{1}{\sqrt{2(N/n-1)}}. \tag{4}$$

Equation (4) can be used to determine the estimation accuracy of the Allan variance method. For a real sample data with finite length, the confidence of Allan variance estimation improves as the number K of independent clusters increases or the averaging time τ decreases. Thus, it is guaranteed that the characteristics of stochastic errors can be obtained with the desired accuracy. For example:

$$\text{if } \sigma(\delta_{AV}) \le 50\%, \text{ there is } n_{MAX} \le N/3 \text{ or } K_{MIN} \ge 3;$$

$$\text{if } \sigma(\delta_{AV}) \le 25\%, \text{ there is } n_{MAX} \le N/9 \text{ or } K_{MIN} \ge 9.$$

Figure 1 Typical structure of Allan characteristic curve



III. MEMS IMU performance

A. IMU characterization

The IMU used in this paper is the Nano IMU (nIMU), a small-size and low-weight MEMS unit, which is made by MEMSense and costs about $1,300, as shown in Figure 2.

The nIMU is compensated for temperature sensitivities to bias and scale factor, and provides serial outputs of 3D acceleration, 3D angular rate and 3D magnetic field data. The key manufacturer specifications of nIMU are listed in Table I (MEMSense, 2009).

B. Test environment

When communicating with nIMU, it is enough to simply connect nIMU to the USB interface board and then connect the USB interface board to the computer. The software used

Figure 2 MEMSense nIMU

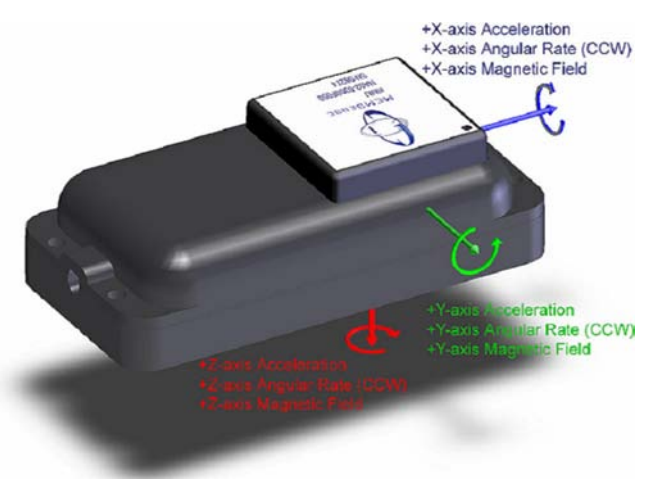


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Table I Key specifications of MEMSense nIMU

Mass (g)	20
Size (mm)	45 × 23 × 13
Operating temperature (°C)	0 to +70
Gyroscopes	
Range (deg/s)	±600
Offset (deg/s)	±1.5
Noise (deg/s)	0.56 (0.95)
Bandwidth (Hz)	50
Accelerometers	
Range (g)	±5
Offset (mg)	±30
Noise (mg)	1.1 (1.3)
Bandwidth (Hz)	50

for acquiring and storing data is the MEMSense IMU Data Console (IDC), which is a console-based, menu-driven application. The MEMSense IDC allows basic data display and collection using the RS422 protocol we specified when ordering the nIMU. Although a relatively simple application displaying as the Main Menu in a DOS window shown in Figure 3, it enables quick and easy verification of device functionality and data collection.

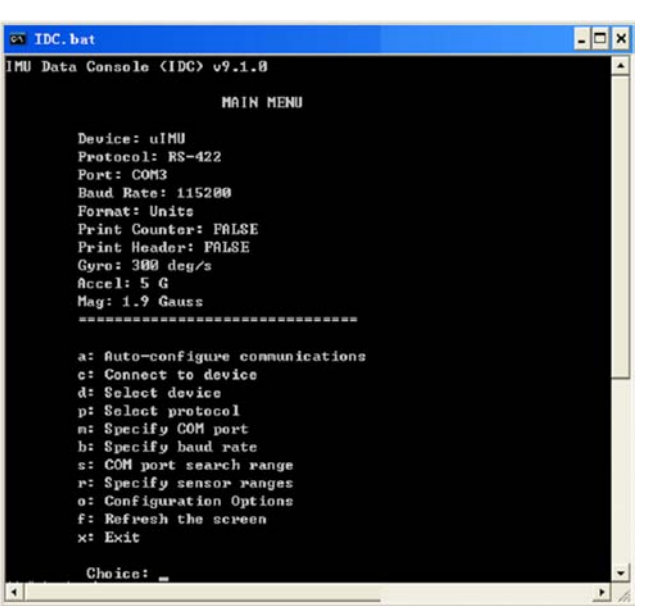
C. IMU error analysis

Normally, the navigation errors when using IMU can be divided into two parts. One is the orientation error, and the other is the distance error. To reduce the overall navigation error, orientation error is the main source of concern. Once the orientation errors occur they will accumulate and grow without bound into unacceptable lateral navigation errors, which can easily lead to a complete failure of the navigation mission. This is another reason for which the error analysis in this paper emphasizes the effect of angular measurement errors produced by the gyroscopes.

We characterize the errors of gyroscopes into two parts: deterministic errors and stochastic errors. The deterministic errors mainly include acceleration and temperature dependent bias offset, the nonlinearity of the scale factor and the scale factor sensitivity to changes in ambient temperature. The stochastic errors mainly include static bias drift, bias instability (BI) and angle random walk (ARW). These errors are considered as the most significant ones among the different types of error sources for consumer grade MEMS gyroscopes.

The deterministic errors can be modeled by calibration and therefore compensated by applying an appropriate set of polynomial compensation functions. The polynomial functions can be of second or third order in practical application, which are determined by considering the error models and the computational burden. However, the stochastic errors may not be directly removed by any compensation schemes rooted in characterization of the deterministic errors; they can only be modeled by a stochastic process. Therefore, the understanding of stochastic errors is of significant importance for the improvement of navigation systems. The goal of this paper is to quickly and accurately identify different noise terms imposed by the stochastic fluctuations using the Allan variance method.

Figure 3 MEMSense IMU data console



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IV. Data acquisition and analysis

The purpose of the experiment conducted in this paper is to identify noise terms existing in MEMS gyroscopes. The bandwidth of MEMSense gyroscopes is 50 Hz as mentioned in the manufacturer specifications. According to the requirement of Nyquist sampling theorem, the sampling rate should be theoretically at least twice the maximum frequency of the sensor. Lots of experiments have proved that the sampling rate should reach three to six times the bandwidth of the sensor for a reliable Allan variance analysis result. In this paper, the data sampling rate is 150 Hz, which is three times the nIMU bandwidth.

In general, the Allan analysis involves collecting static data for a time period ten times longer than the most significant noise correlation time (i.e. the averaging time). For most MEMS gyroscopes, the correlation time ranges from 1 second to 1 hour or even longer time periods. In our research, the experiment was implemented at room temperature and the measurements were collected from nIMU during each 6 hour static test for seven days.

The distribution histogram of the static data of the *X*-axis gyroscope is shown in Figure 4, which indicates that the data distribution is similar to a normal distribution. A normplot of the sample data is shown in Figure 5, and it is obvious that the static data distribution is a normal distribution.

The average of sample data from each axis of the gyroscope is the static bias drift. Although significant, this error is not the focus of error analysis and error model, because a trivial calibration method can effectively compensate for it. The static bias drifts of each axis during the seven-day test are calculated and listed in Table II. It can be seen that MEMS gyroscopes have large bias drifts, which can accumulate several degrees of orientation error after even 1 min. Thus, the static bias drift must be removed from the raw sample data. Once the static bias error is compensated, other sources of stochastic errors will become dominant.

Furthermore, there is a warm-up period where the sensor's internal temperature increases from initial value to operating

Figure 4 Static data distribution compared to the normal distribution

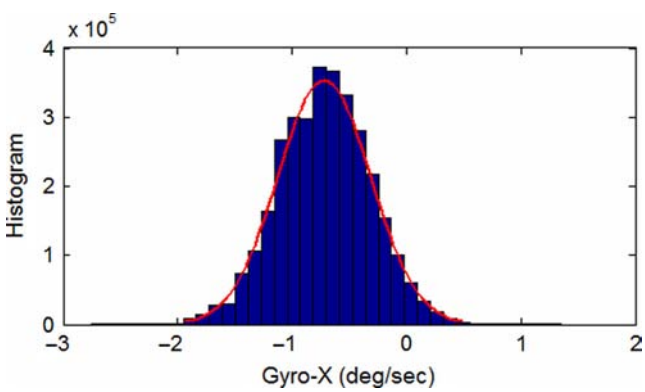


Figure 5 Normplot for the *X*-axis static data

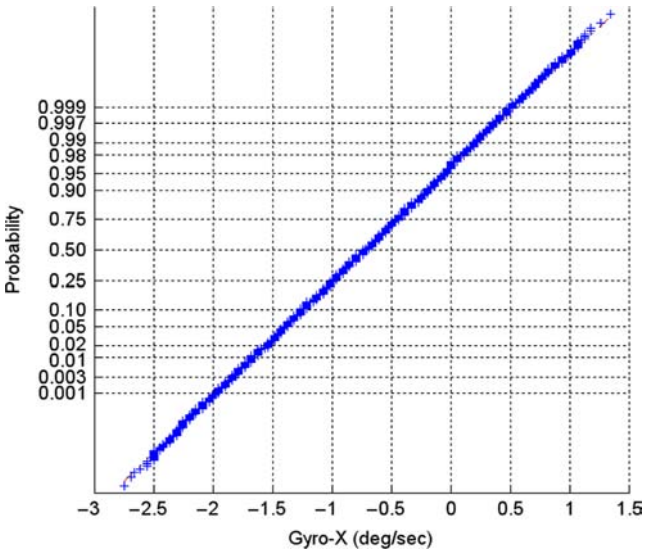


Table II Static bias drifts for the seven-day test (deg/s)

	D1	D2	D3	D4	D5	D6	D7
Gyro-X	−0.7186	−0.6539	−0.6160	−0.5641	−0.4988	−0.5120	−0.4945
Gyro-Y	0.6315	0.5964	0.5991	0.5703	0.5588	0.4847	0.5424
Gyro-Z	−0.6457	−0.6236	−0.5833	−0.5207	−0.5136	−0.4660	−0.4834

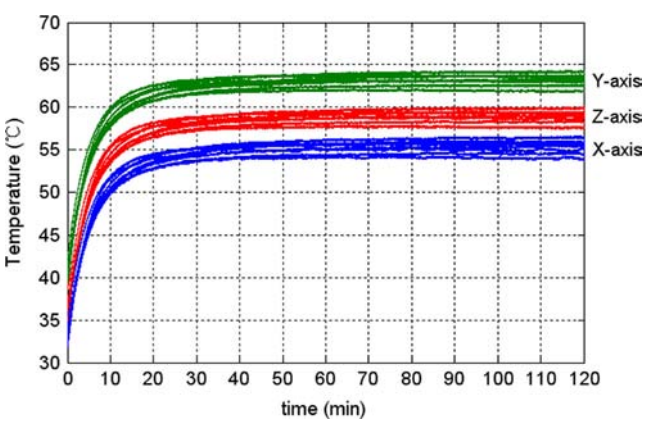
value, as shown in Figure 6 for the seven-day test. It can be seen that the warm-up phase of nIMU lasts approximately 1 hour. Therefore, the sensors have to be warmed up for at least 1 hour before data acquisition so as to avoid any temperature dependent drift.

V. Allan variance analysis result

The basic procedure of the normal Allan variance method can be summarized as follows:

- *Step 1.* Take a long sequence of data and divide it into clusters according to the averaging time.
- *Step 2.* Average the data in each cluster.
- *Step 3.* Calculate the differences in average between successive clusters, square these differences, add them all up, and then divide the sum by a rescaling factor to get the Allan variance.
- *Step 4.* Obtain the Allan standard deviation by extracting the square root of the Allan variance.
- *Step 5.* Go back to step 1, increase the averaging time, and then start over again until the data can only be divided into the allowable minimum number of independent clusters.

Figure 6 Internal temperature profile of the warm-up period



The six-hour static sample data can be used to characterize the overall noise of nIMU through the Allan variance method, and this will be detailed as follows.

A. Performance of the normal Allan variance method

During the normal Allan variance analysis process, the entire six-hour sample data are divided into clusters including consecutive data points, which means that the cluster size starts at 1 and increases by 1, namely $n = 1, 2, 3, \ldots, n_{MAX}$. The entire data are first analyzed using the normal Allan variance method, and the log-log plot of $\sigma(\tau)$ versus τ for the triaxial gyroscope is shown in Figure 7.

Inspection of the curves shows that the Allan variance result reveals two observable random noise components significant to nIMU gyroscope: BI and ARW. It is clear that the ARW is the dominant noise term for the shorter averaging times (less than 10) whereas the BI is the dominant noise term for the long averaging times (hundreds of seconds).

The noise coefficients can be acquired from the Allan variance analysis result and graphically demonstrated in the log-log plot. A straight line with slope of $-1/2$ is fitted to the shorter averaging time region of the plot. The noise coefficient Q of ARW can be obtained by reading the slope line value σ_a at $\tau = 1$ (Point A in Figure 7). Since the unit of ARW is $\text{deg}/\sqrt{hr}$, there is $Q = \sigma_a/60$. The almost flat part at the bottom of the curve in the long averaging time region is fitted by a zero slope line. The noise coefficient R of BI can be obtained by reading the point value σ_b where the zero slope line meets the ordinate axis (Point B in Figure 7), and there is $R = 0.6648\sigma_b$. The identified noise coefficients for nIMU triaxial gyroscope are listed in Table III.

For the six-hour static test in this paper, the total number of data points is $N = 3{,}240{,}000$, and the minimum number of independent clusters is set to $K_{MIN} = 9$, which indicates that the largest estimation percentage error is about 25 percent according to equation (4). Using the normal Allan analysis

Figure 7 Allan variance analysis result of nIMU gyroscope

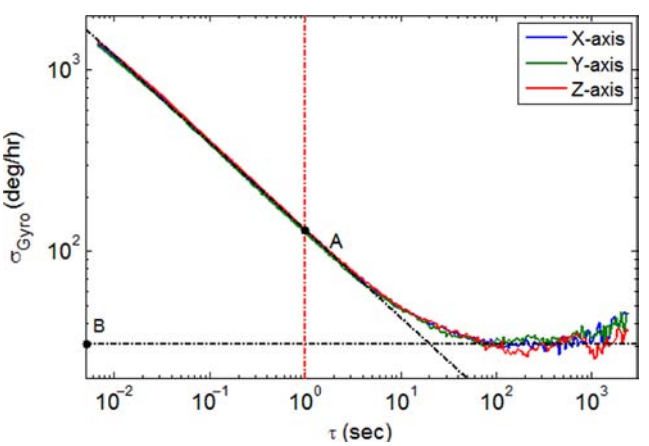


Table III Noise coefficients for nIMU gyroscope

	$Q(\text{deg}/\sqrt{hr})$	$R(\text{deg}/\sqrt{hr})$
X-axis	2.1282	46.3014
Y-axis	2.0992	48.3814
Z-axis	2.1983	43.9352

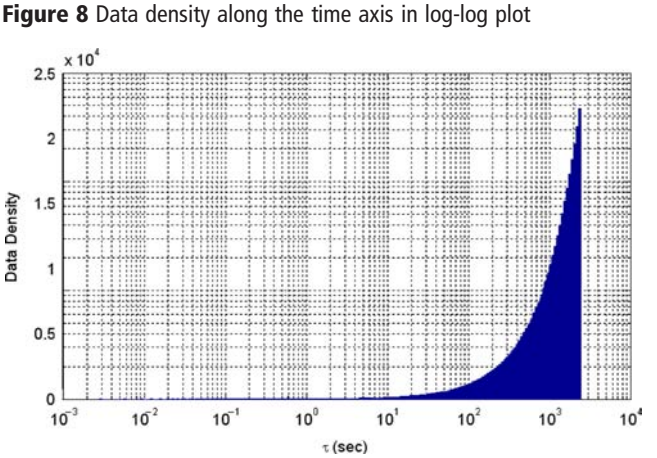
method, 36,000 Allan standard deviations need to be calculated according to the ever-increasing cluster size. As a result, the deviation computation time is unacceptable when using the MATLAB software, which is approximately four days when running on a standard 1.83 GHz Windows XP Desktop with 1G memory, three days when running on a standard 2.2 GHz Windows XP Desktop with 2G memory and 30 hours when running on a standard 2.8 GHz Windows 7 Desktop with 4G memory. Hence, the normal Allan variance method has the shortcomings of having huge working set, heavy system load, and therefore needs long computation time. In order to reduce the calculation burden and speed up the estimation process, the normal Allan variance method needs to be improved while keeping the accuracy of the results. By default, the computation time mentioned in the rest of this paper is the running time on the 2.2 GHz Windows XP Desktop with 2G memory.

B. Performance of the modified Allan variance method

As mentioned above, the cluster sizes form an arithmetic sequence in the normal Allan analysis method, and so do the averaging times in the form of $\tau = T_{MIN}, 2T_{MIN}, \ldots, n_{MAX}T_{MIN}$, where T_{MIN} is the minimum averaging time and also the regular common difference. Throughout this paper, there is $T_{MIN} = T_s$. Since the Allan standard deviation is plotted on a log-log scale for the convenience of intuitively analyzing the different stochastic noise terms, the time axis is a logarithmic coordinate. As shown in Figure 8, this results in the data in the lower right part of the Allan characteristic curve being more and more dense than the data in the upper left part.

Various strategies have been tried by researchers to save the computation time, such as enlarging the minimum averaging time T_{MIN} with additional averaging operation or forming the cluster sizes into geometric sequences with integral common ratios (Allan, 1987; Riley *et al.*, 2008). In some works, T_{MIN} has a value of 1 s or other multiples of T_s. Based on the discussion in Section II, the Allan deviation curve of low-accuracy inertial sensor is closer to the deviation axis than that of the high-accuracy inertial sensor. The above experiment indicates that the ARW is the dominant noise term of nIMU gyroscope for the averaging times less than 10 s. Thus, in order to accurately estimate the noise coefficient of ARW existing in the low-accuracy inertial sensor, the value of T_{MIN} cannot be too large. In some other works, the cluster size is suggested to be $n = 2^l (l = 0, 1, 2, \ldots, l_{MAX})$. For the six-hour static test, under the condition of $N = 3{,}240{,}000$ and $K_{MIN} = 9$, if $n = 2^l \leq 360{,}000$, there is $l_{MAX} = 18$. Therefore, only 18 Allan standard deviations need to be calculated. It is generally thought that too few deviations would not adequately represent the overall trend of the Allan characteristic curve.

Figure 8 Data density along the time axis in log-log plot



In practical application, there is no need to calculate the Allan standard deviations for as many averaging times as possible, which means that the cluster sizes do not need to be consecutive as in the normal Allan variance method. If the data in the lower right part of the curve (i.e. the long averaging time region) could be appropriately reduced, and the obtained results can still represent the overall trend of the Allan characteristic curve, the calculation burden will be reduced while the exactness of the analysis result can be ensured.

According to the features of the log-log plot, if the Allan deviation data need to be evenly distributed along the logarithmic-scale time axis, the averaging times should form a geometric sequence, namely $\tau = T_{MIN}, T_{MIN}T_r, \ldots, T_{MIN}T_r^{i_{MAX}-1}$, where T_r is the common ratio, i_{MAX} is the maximum number of Allan standard deviations that need to be calculated and there exists $T_{MIN}T_r^{i_{MAX}-1} = n_{MAX}T_s = T_{MAX}$. Therefore, based on the character of the normal Allan variance method, a strategy of transforming the averaging time series from an arithmetic sequence into a geometric sequence is proposed in this paper.

Following the ideas formulated above, the basic procedure of the modified Allan variance method is summarized as follows:

- *Step 1.* Calculate the allowable value of T_{MIN} and T_{MAX} of the averaging time based on the sampling rate and the required estimation accuracy.
- *Step 2.* Determine the number i_{MAX} of the Allan standard deviations according to the desired estimation time.
- *Step 3.* Compute the common ratio T_r so as to yield the initial geometric sequence of averaging times.
- *Step 4.* Obtain the valid (i.e. round to integers and remove duplications) cluster sizes corresponding to the averaging time series.
- *Step 5.* Get the final feasible geometric sequence of averaging times based on the valid cluster sizes.
- *Step 6.* Calculate the Allan standard deviations versus the obtained averaging time series using the normal Allan variance method.

For the same 6 hour static test with $N = 3{,}240{,}000$ and $K_{MIN} = 9$, the characteristic curve obtained by the modified Allan variance method with $i_{MAX} = 300$ is compared to that obtained by the normal Allan variance method, as shown in Figure 9. The comparison result shows that the two curves basically coincide with each other in the shorter averaging time region, and even in the long averaging time region the red curve can still represent the overall trend of the characteristic curve obtained by the normal Allan variance method.

Using the modified time-controllable Allan variance method, the identified noise coefficients for nIMU triaxial gyroscope are listed in Table IV.

Figure 9 Comparison of the two characteristic curves

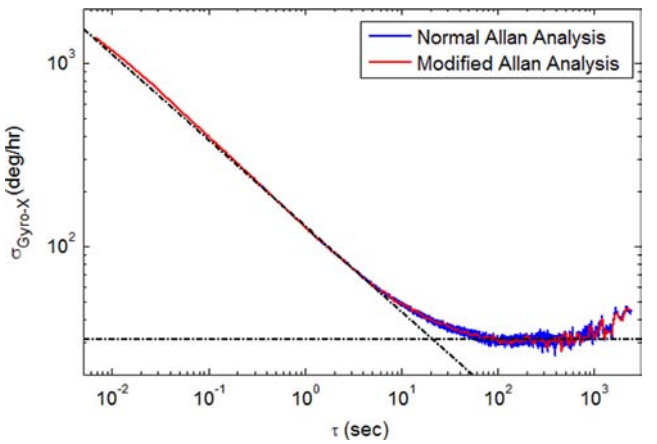


Table IV Noise coefficients for nIMU gyroscope when $i_{MAX} = 300$

	$Q(\text{deg}/\sqrt{hr})$	$R(\text{deg}/\sqrt{hr})$
***X*-axis**	2.1286	46.0565
***Y*-axis**	2.0980	48.8889
***Z*-axis**	2.2014	43.8061

In this situation, the deviation computation time is only about 80 seconds, which indicates that the modified Allan variance method can speed up the estimation process enormously. Note that the number of calculated Allan standard deviations is not fixed in the modified method, which makes the computation time controllable when choosing different values of i_{MAX}. This is more convenient in practical applications, especially for long-term sample data. However, reducing the amount of Allan standard deviations is thought to decrease the estimation accuracy. Thus, the performance of the modified time-controllable Allan variance method needs to be discussed in the following section.

VI. Discussion

A. Accuracy loss of the modified Allan variance method

In order to evaluate the performance of the modified Allan variance method, the estimation results obtained by the normal Allan variance method are used to serve as the relative truth value. The relative accuracy loss of noise coefficient estimation is defined as:

$$E_{Allan} = \frac{|C_{Normal} - C_{Modified}|}{C_{Normal}}, \tag{5}$$

where C_{Normal} is the noise coefficient obtained by the normal Allan analysis, and $C_{Modified}$ is the noise coefficient obtained by the modified Allan analysis.

Based on formula (5), when $i_{MAX} = 300$, the relative accuracy losses of noise coefficients estimation for nIMU triaxial gyroscope are given in Table V, where $E_{Allan-ARW}$ is the relative accuracy loss of ARW estimation and $E_{Allan-BI}$ is the relative accuracy loss of BI estimation. The result shows that the relative loss of noise estimation accuracy is very small or even negligible, especially for the ARW estimation.

As mentioned above, the computation time of the modified Allan variance method is controllable by limiting the value of i_{MAX}. We define T_C to be the computation time measured in seconds according to different i_{MAX}. The evaluation results for X-axis gyroscope are listed in Table VI.

Table V Relative loss of noise estimation accuracy for nIMU gyroscope

	$E_{Allan-ARW}$ (%)	$E_{Allan-BI}$ (%)
***X*-axis**	0.0217	0.5287
***Y*-axis**	0.0581	1.0489
***Z*-axis**	0.0426	0.2939

The test results in Table VI have the following features:

- $E_{Allan-ARW}$ is the minor part of accuracy loss involved in the modified Allan variance method, whose maximum value is less than 0.08 percent and minimum value is almost zero since the situation of $i_{MAX} = 1{,}800$.
- $E_{Allan-BI}$ is the major part of accuracy loss involved in the modified Allan variance method, whose maximum value is less than 0.53 percent and minimum value is greater than 0.03 percent.
- It is not always true that the more Allan standard deviations, the closer the estimation result is to the relative true value. For example, the value of $E_{Allan-BI}$ in the situation of $i_{MAX} = 2{,}700$ is greater than that of $i_{MAX} = 2{,}400$, and the value of $E_{Allan-ARW}$ in the situation of $i_{MAX} = 600$ and $i_{MAX} = 900$ is greater than that of $i_{MAX} = 300$.

The above discussion can be summarized as follows:

- The computation time T_C can be adjusted by varying the value of i_{MAX}.
- $E_{Allan-ARW}$ is relatively small and can be ignored compared to $E_{Allan-BI}$. The noise coefficients of ARW obtained by the normal and the modified Allan variance methods, respectively, become the same with no estimation accuracy loss since $i_{MAX} = 1{,}800$.
- Although there exists accuracy loss of noise coefficient estimations, the modified Allan variance method can dramatically speed up the calculation with the standard deviations evenly distributed along the logarithmic-scale time axis. Even for the situation of $i_{MAX} = 3{,}000$, there is $T_C = 600$ s (i.e. about 10 min), which is still acceptable in practical applications.

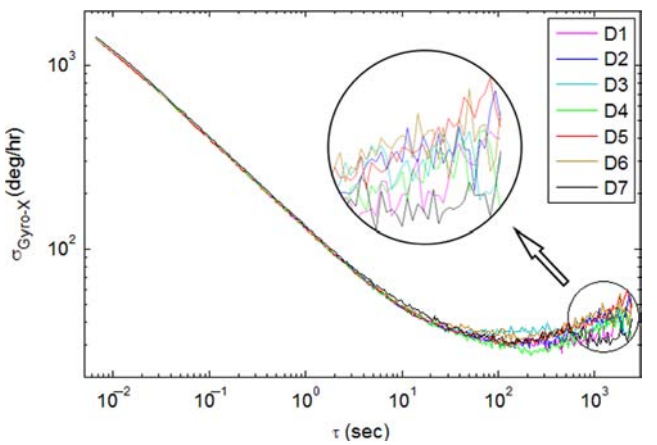
B. Feasibility of the modified Allan variance method

Based on the seven-day static test of nIMU, the Allan variance results for X-axis gyroscope with $i_{MAX} = 300$ are shown in Figure 10.

Table VI Evaluation of the time-controllable Allan variance method

i_{MAX}	i'_{MAX}	T_C (s)	$E_{Allan-ARW}$ (%)	$E_{Allan-BI}$ (%)
300	249	75	0.0217	*0.5287*
600	466	145	*0.0773*	0.5235
900	670	225	0.0552	0.2224
1,200	867	270	0.0046	0.2219
1,500	1,058	320	0.0063	0.1426
1,800	1,244	375	*0.0000*	0.1065
2,100	1,426	430	*0.0000*	0.0867
2,400	1,605	480	*0.0000*	*0.0336*
2,700	1,781	540	*0.0000*	0.1073
3,000	1,954	600	*0.0000*	0.0697

Figure 10 Allan variance results for *X*-axis gyroscope with $i_{MAX} = 300$



The results show that the seven curves essentially coincide with each other in the shorter averaging time region, but show different characteristics in the long averaging time region. The reasons for this phenomenon can be explained as follows:

- The ARW is the dominant noise term in the short averaging time region where the number of independent clusters is very large. Therefore, the confidence of the noise coefficient estimation is sufficiently high, which results in the seven curves essentially coinciding with each other. Even for the averaging time as long as $\tau = 10$, according to equation (4), the percentage error is only 1.5218 percent. In fact, the line with slope of $-1/2$ is only fitted to the time region from 0.05 to 1 s, thus the maximum percentage error can be further reduced to 0.4811 percent, and the magnitude of the estimation error can be calculated as $E(\delta_{AV-ARW}) = Q \times 0.4811$ percent.
- The BI is the dominant noise term in the long averaging time region where the number of independent clusters is decreasing, and as well as the confidence of the noise coefficient estimation, which results in the difference between the characteristic curves becoming more and more obvious. In fact, the zero slope line is fitted to the time region from 100 to 500 seconds, according to equation (4), the percentage error in this region can reach to 10.885 percent, and the magnitude of estimation error can be calculated as $E(\delta_{AV-BI}) = R \times 10.885$ percent.

The estimated noise coefficients for the nIMU triaxial gyroscope during the seven-day test are listed in Table VII. It can be seen that the standard deviations of the identified noise coefficients for ARW are negligible compared to those of the BI, and the standard deviations of all the identified noise coefficients are either smaller than or close to the Allan estimation error. The estimation results clearly show the feasibility and validity of the modified time-controllable Allan variance method, and clearly indicate that the random noise terms of nIMU triaxial gyroscope have relatively good repeatability.

Table VII Noise coefficients for nIMU gyroscope based on the seven-day test

	X-axis	*Y*-axis	*Z*-axis	*X*-axis	*Y*-axis	*Z*-axis
		$Q(\text{deg}/\sqrt{hr})$			$R(\text{deg}/\sqrt{hr})$	
D1	2.1286	2.0980	2.2014	46.0565	48.8889	43.8061
D2	2.1576	2.0961	2.2025	51.2393	49.4632	39.7205
D3	2.1486	2.1095	2.1952	46.6847	49.6319	49.4737
D4	2.1420	2.1081	2.1975	53.3238	51.3009	45.1676
D5	2.1417	2.1145	2.1848	43.5516	46.3863	45.5255
D6	2.1327	2.0852	2.1957	48.4844	49.4174	45.7291
D7	2.1317	2.0965	2.2422	51.1595	45.5233	49.2851
STD	0.0103	0.0101	0.0183	3.4510	2.0083	3.3303
		$E(\delta_{AV-ARW})$			$E(\delta_{AV-BI})$	
MAX	0.0104	0.0102	0.0108	5.8043	5.5841	5.8154
MIN	0.0102	0.0100	0.0105	4.7406	4.9552	4.7683

C. Limitations of the modified Allan variance method

Although the modified Allan variance method is useful for characterizing the stochastic noise terms of MEMS IMU and reducing the calculation burden of normal Allan variance method, it still has some limitations:

- The modified and normal Allan variance methods do not always determine a unique noise spectrum, because the mapping from the spectrum to the Allan variance is not one-to-one. In general, the result does not completely characterize the covariance properties of the noise.
- The basic idea of the modified Allan variance method is to form the cluster sizes into geometric sequences, so that the Allan deviation data can be evenly distributed along the logarithmic-scale time axis. However, reducing the amount of Allan standard deviations can accordingly affect the estimation accuracy to a certain extent.
- Since the main effect of the modified Allan variance method is a significant decrease of the deviation data in the long averaging time region where the BI is the dominant noise term, the influence on the estimation accuracy of BI is larger than that on ARW.
- In the process of the modified Allan variance analysis, the actual number of calculated Allan standard deviations is reduced to less than the desired number i_{MAX} when the cluster sizes are rounded to their nearest integers (with the duplications removed).

Future research will further improve the limitations of the modified Allan variance method and put the identified noise coefficients into practical use to verify the feasibility and validity of the modified method. Furthermore, the modified Allan variance method will be used to identify noise terms existing in different grade IMUs, and thereby enhance the understanding of its performance.

VII. Conclusion

The Allan variance method is a simple and efficient time-domain method, which can provide information on the types and magnitudes of various stochastic noise terms contained in the inertial sensor data. Based on the disadvantages of the normal Allan variance method and the features of data distribution in the log-log plot of the Allan standard deviation versus the averaging time, a modified time-controllable Allan variance method is proposed in this paper, which can greatly reduce the calculation burden and speed up the

estimation process with the computation time controllable while ensuring the accuracy of estimation. A strategy of transforming the averaging time series from an arithmetic sequence into a geometric sequence is adopted in the modified Allan variance method, thus the Allan standard deviations can be evenly distributed along the logarithmic-scale time axis in the log-log plot.

A series of six-hour static tests have been implemented at room temperature and the static measurements have been collected from the MEMSense nIMU. The results show that the computation time and estimation accuracy are acceptable in practical applications, which proves the feasibility and validity of the modified Allan variance method proposed in this paper. The test results also indicate that the random noise terms of nIMU triaxial gyroscope have relatively good repeatability and each individual test result can be applied for system prediction and analysis.

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